Inequality in a Potential Equalizer of Opportunity*

Variation in the Accessibility and Net Benefits of English-Medium Instruction in India

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This document contains the Introduction and Figure 3 of the titular paper. The project may have been updated since this file was generated; the most up-to-date versions of this document, the full paper, the replication package, and other application samples are available online for those interested. This PDF can be generated by running `make samples` or `scripts/run_full_build.sh` from the repo root.

Abstract

English-medium instruction (EMI), or the teaching of school subjects in English, is often viewed as a potential palliative for the intergenerational rigidity of socioeconomic status in India, where English skills command substantial wage premia. I study whether the conditions governing households' access to and potential net gains from EMI are themselves unequally distributed. I organize my analysis around three such conditions: inherited linguistic costs to acquiring English, access to English-medium schooling, and the local economic setting in which English skills may be rewarded. By harmonizing district data on languages, schooling, consumption, migration, human capital, and sectoral employment, I find inequality across all three. Inherited linguistic conditions are highly persistent and predict future human capital gains, financial asset access, and structural change, but their relationship with language acquisition behavior varies across institutional settings. EMI exposure and provision are strongly organized by schooling system and sharply stratified along socioeconomic lines, even within the same local education market. Consumption growth is more associated with EMI enrollment than enrollment alone, and the strength of their association increases in pre-existing human capital and urbanization. In sum, a high-return educational opportunity need not equalize opportunity when access to it and the resources which help benefit from it are unequally distributed.

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1 Introduction

In an Indian economy characterized by rapid GDP growth alongside wage stagnation, profuse informality, and high rates of both youth unemployment and underemployment (International Labour Organization 2024), what can a household do to improve the future prospects of their children? The history of economics highlights *education* as one answer (Mincer 1974; Card 1999; Björklund and Kjellström 2002; Bhandari and Bordoloi 2006). The history of India highlights another: *English*. In a context of extraordinary linguistic diversity (Gurevich et al. 2025) and thus extraordinary communication and coordination costs (Nakagawa and Sugasawa 2021), English has come to serve as a primary domestic lingua franca (Clingingsmith 2014). In its capacity as a global lingua franca (Melitz 2016), meanwhile, English opens pathways both internationally—e.g., to education at prestigious universities or employment at multinational firms (Ramanathan 2014; Bedi 2019, 32–33)—and at home. The domestic expansion of internationally traded services and information technology, in particular, has led to a flurry of jobs whose higher socioeconomic returns (Chamarbagwala and Sharma 2011; Smokotin et al. 2014) are largely accessible only to those who can use English (Mankiw and Swagel 2006; Shastry 2012).

Empirical study corroborates the notion that English expands economic mobility (Itani et al. 2015). Using nationally representative household data, Azam et al. (2013) find that English fluency was, at least up until 2005, associated with a 34% hourly wage premium for men and a 22% premium for women; even limited English proficiency was associated with premia of 13% and 10%, respectively. Refeque and Azad (2022) find similar differentials extending through 2011–12. Evidence for causality in this association comes from Chakraborty and Bakshi (2016), who interpret West Bengal's 1983–2004 ban on English teaching in public primary schools as a natural experiment where cohorts more *exposed* to the policy earned lower subsequent wages. "Language skills are human capital," after all (Chiswick and Miller 1995, 4), and in a context like India especially, "English-language skills [specifically] are a form of human capital" (Azam et al. 2013, 344).

It is thus by combining our two answers—education and English—that we can begin to understand the demand for *English-medium instruction (EMI)*, or for the teaching of all school subjects in English, currently seen in India (Gooptu and Mukherjee 2023; Lahoti and Mukhopadhyay 2019; Chattopadhay and Roy 2017). For its potential to expand equality of opportunity, EMI has been touted as a tool of social development (National Knowledge Commission 2009) and described as a route to escape caste (Mathur 2013; Mathew and Srivastava 2011; Vij 2020) alongside the rigid wage, employment, and occupation distributions it enforces across generations (Madheswaran and Singhari 2016; Munshi and Rosenzweig 2006).

But it is ultimately households who decide whether or not a child attends EMI, and the determinants of their choice—the costs they face, the benefits they expect, and the forces constraining the choice set in front of them—are affected by factors which are not just national and historical, but also specific and local. My focus is on whether

such factors are associated with the same inequalities that English skills are often expected to help households escape. I study inequalities which exist along dimensions such as social group, religion, parental education, household resources, and urban or rural residence. If the ability to access or benefit from EMI is systematically related to these characteristics, even within the same local education market, then we have evidence that a high economic payoff to English skills alone is not enough for EMI to become an equalizer of opportunity.

Though I never estimate a structural model of household choice, their investment problem still provides a helpful organizing framework. I will first study a type of cost whose variation is predetermined relative to the institutional and economic factors that otherwise affect the household's problem: **inherited linguistic costs to learning English**. These will reflect historically persistent features of the local linguistic environment (see Rowe and Weisleder 2020) which affect the opportunity cost of learning English over other domestic bridge languages, namely Hindi. This work, detailed in Section 3, owes much credit to Shastry (2012); in line with her methodology, I create several different measures of local linguistic distance from *Hindi* to capture this opportunity cost. In Section 4 I analyze the **accessibility of English-medium instruction** through enrollment, various measures of EMI exposure, and private school management. Finally, in Section 5 I study potential determinants of **expected returns to English skills**. My goal here is to find evidence on the value of English skills in the local labor market, with the assumption that households incorporate the local history of such returns into their beliefs regarding English, akin to households having adaptive expectations. I ask whether EMI exposure predicts later district consumption and whether that association is stronger where pre-existing human capital and urbanization are greater. I then examine whether inherited linguistic conditions accompany human-capital growth and skilled in-migration, as well as changes in financial access and economic structure.

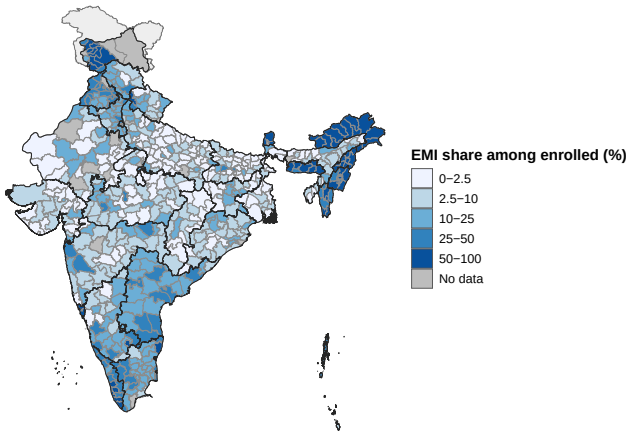
The local linguistic environment, the local education market, and the local labor market—all three of the factors I focus on are studied locally. These choices are justifiable: the linguistic conditions which households pass down to their children are highly persistent at the local level, local schooling conditions constrain the menu of educational opportunities households choose from, and local economic conditions plausibly cause households to update their beliefs on the economic returns to learning English. I will use a household's district to proxy for their 'local' environment.[1] Though the district is the most granular unit at which I can harmonize the longitudinal data sources I rely on, it must be noted that interdistrict migration would confound and attenuate any association between the characteristics of one district's education market and its future economic outcomes. Older studies found interdistrict migration rates (and particularly rates of economically-motivated interdistrict migration) to be so low even after the expansion of IT and services in the 1990s (Topalova 2005; National Statistical Office 2021) that they "assume[d] that district of current residence (or of employment) of an individual is approximately

[1]A district in India is a political unit akin to a county in the United States, albeit with an average population of around 2 million. Table 1 contains summary statistics derived from the distribution of district populations.

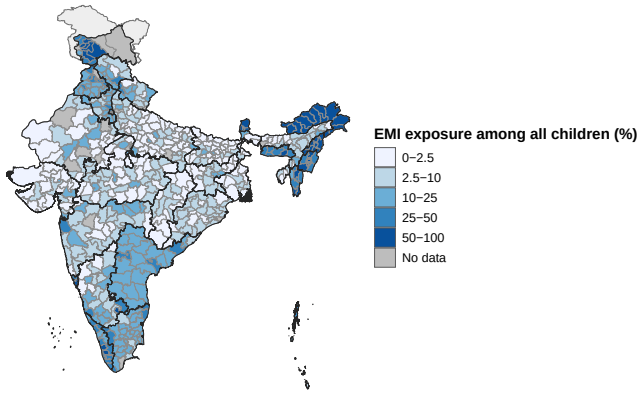
the same as the schooling district" (Chakraborty and Bakshi 2016, 6). More recent work, however, complicates (Mahapatro 2020; Rajan et al. 2020) and even outright repudiates (Lucas 2021) this notion. These studies emphasize the role of the IT expansion in driving interdistrict migration (Economic Divison 2015; Ghose 2026), and my own results associate potential determinants of household investment into EMI with the skill composition of recent work migrants. I return to those results in Section B. Regardless, it is evident that district outcomes are not an aggregation of individual returns to EMI, but a function of places and their changing populations.

The results show that the conditions surrounding English-skill investment are themselves unequally distributed. The relative cost of acquiring a bridge language is partly inherited, and language behavior varies with these inherited linguistic conditions. Formal English-medium schooling is strongly organized by state boundaries, particularly in the public sector. Within the same district, social-group differences remain large in English-medium and private-school participation, with larger differences in urban schooling markets. EMI exposure is more strongly associated with later local consumption than enrollment alone, and this association is stronger in districts with higher initial human capital and urbanization. The inherited linguistic gradient is also associated with later changes in human capital, the skill composition of recent work migrants, financial access, and sectoral composition.

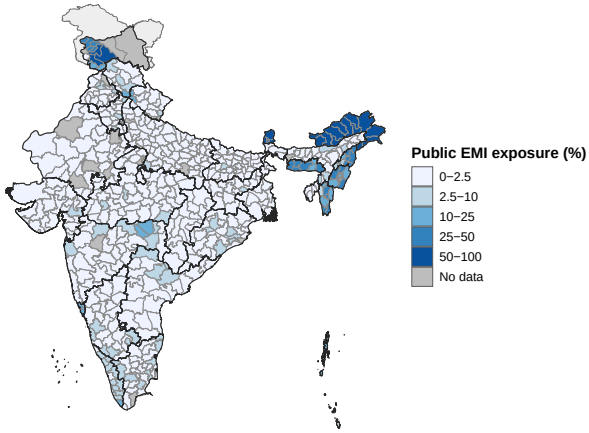
Taken together, the evidence limits the equalizing promise of a high-return educational opportunity. Households begin from unequal linguistic conditions and face unequal access to EMI; among districts, the later consumption association is stronger where baseline human capital and urbanization are higher. English skills can be valuable while access to English-medium schooling and the local conditions associated with its later economic returns remain unequal.



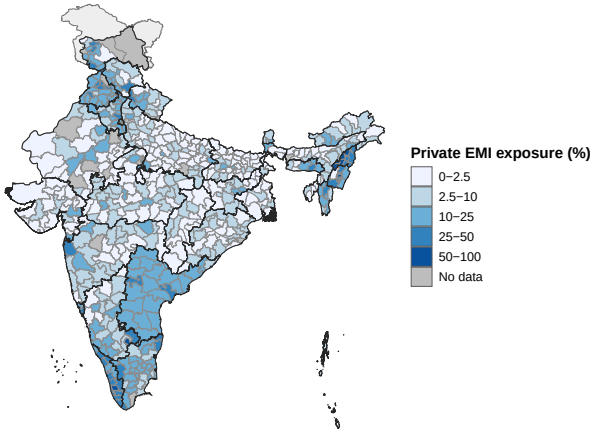
(a) EMI share among enrolled children.



(b) EMI exposure among all children.



(c) Public-school EMI exposure among all children.



(d) Private-school EMI exposure among all children.

Figure 3: District distribution of English-medium schooling measures on 2001 Census boundaries. All panels use a common percentage scale. Source: NSS 64th Round, 2007–08 (National Sample Survey Office 2011). Tengshe (2014) and Natural Earth (2022) provide geospatial data. District boundaries reflect 2001 geography. Politically disputed regions are identified using Natural Earth, and those without data—Aksai Chin, Azad Kashmir, Gilgit-Baltistan, the Trans-Karakoram Tract, and Siachen Glacier—are colored light gray. Their inclusion does not imply a position on territorial disputes.